



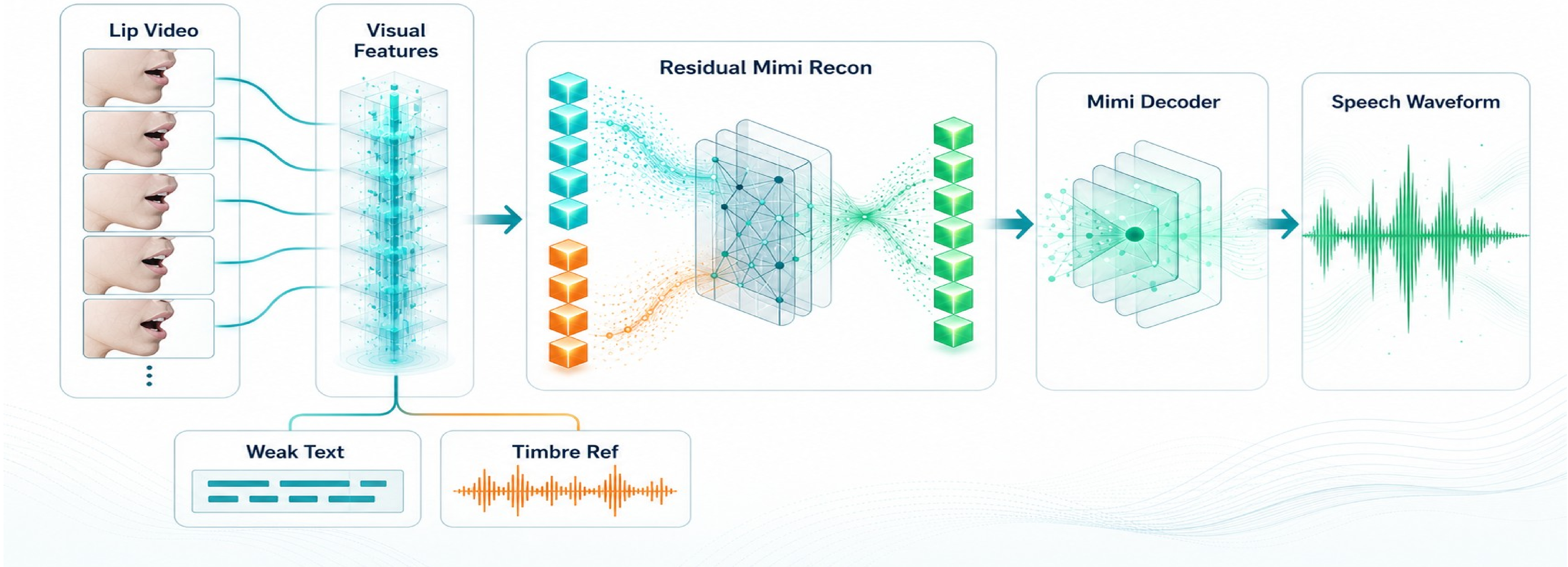
Audio-Centric Visual Speech Reconstruction

Lip and audio latents carry the speech signal; text is a weak alignment cue

Replace vision -> text -> TTS with direct Mimi latent reconstruction

1 SYSTEM ARCHITECTURE

Audio-Centric AVSR



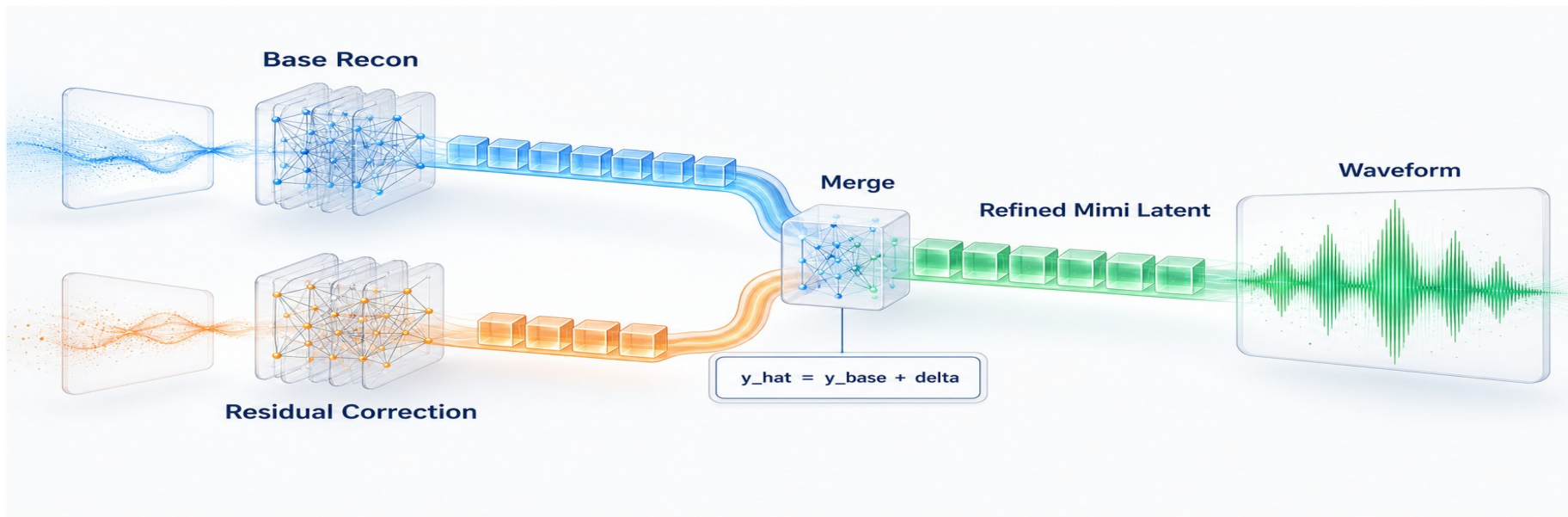
2 DATA / CONDITIONS

lip_avsr.npy	Auto-AVSR lip crop latent
avsr_enc_lipavsr.npy	AVSR visual encoder state
avsr_text_lipavsr.txt	noisy text, weak semantic aid
SmolM2 hidden	aligned text-token hidden states
speaker/timbre	speaker_emb + Mimi prompt stats
target	Mimi audio latent to reconstruct

The final path uses AVSR-compatible lip crops rather than the old RGB lip crop. External videos use lip-AVSR text with uniform alignment.

text is support, not bottleneck

3 DETERMINISTIC RECON

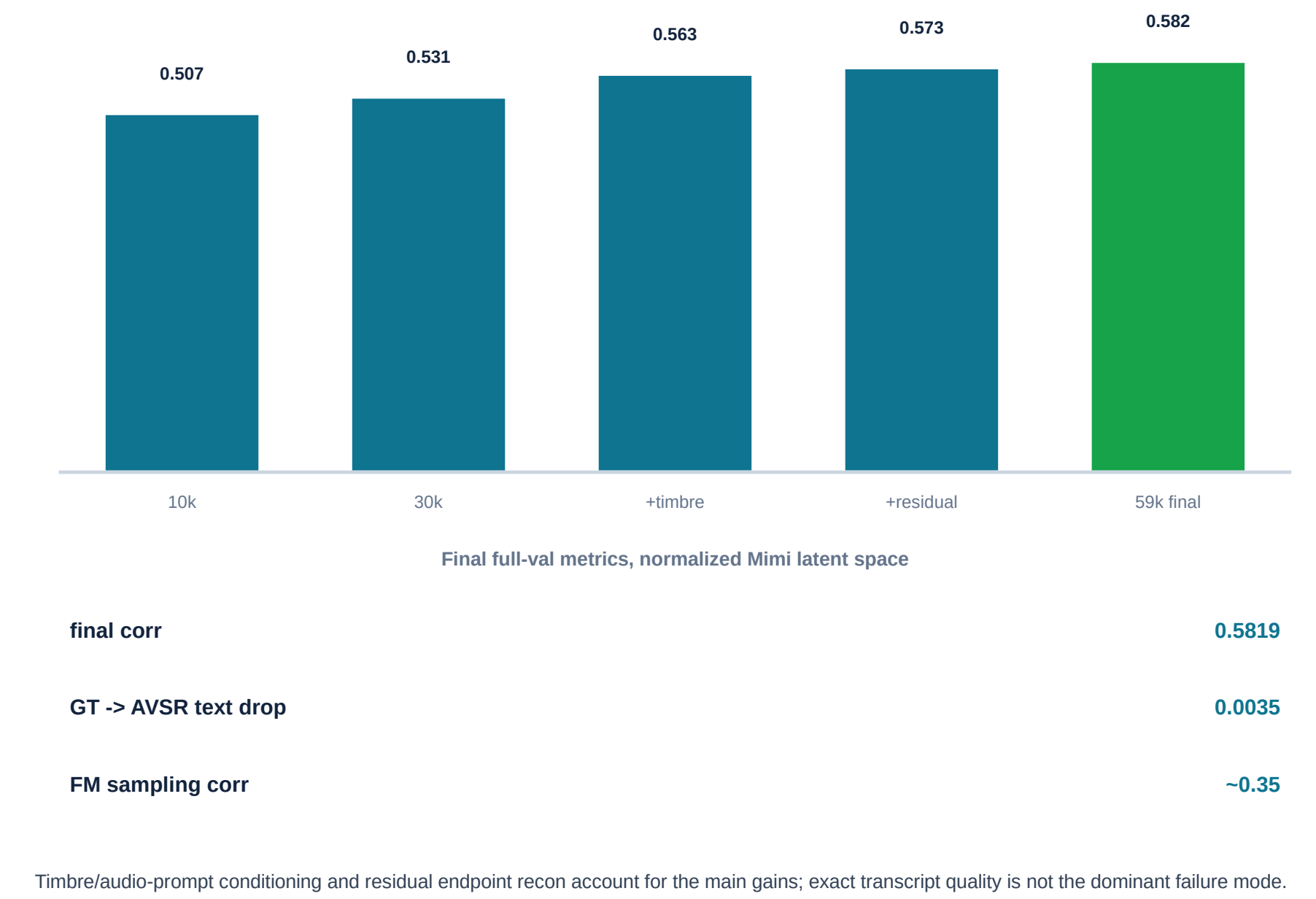


$$C = \{\text{visual, text, speaker, audio prompt, timbre}\}$$

$$\hat{y} = y_{\text{base}} + \delta$$

The final checkpoint fixes $x_{\text{tld}} = 0$ and $\tau = 1$. It is endpoint regression through a frozen base plus residual correction, not an iterative FM sampling loop.

4 EVIDENCE



5 SILENT-REFERENCE DEMO



silent input



reference waveform



generated crop

Trump example: silent MP4 + short reference segment -> voiced MP4. The visual stream is preserved; only the speech track is reconstructed.

6 TAKEAWAY

We should not treat lip video as a text-recognition problem followed by speech synthesis.

Lip motion provides phonetic and timing evidence.
Mimi latents preserve codec-compatible acoustic structure.
Weak text stabilizes content but does not need to be perfect.
Timbre/reference conditions carry speaker and recording color.

This supports an audio-latent reconstruction architecture over a strict text cascade.

7 ARTIFACTS

scripts/run_raw_video_avsr_recon_pipeline.py
scripts/gradio_avsr_gui.py
poster/fm_avsr_poster.pptx and PDF preview
data/assets/trump_silent_ref_demo/*.mp4
eval_out/trump_raw_prompt_pipeline/./avsr_text_lipavsr.txt

Generated visual components are stored under poster/assets/generated_*.png.